

Science — Cbse Periodic Test (Pt) — 40 Marks

ANSWER KEY — FOR GRADING

Subject: Science | Class: Class 9

Total Marks: 40 | Time Allowed: 1h 30m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	An animal is observed in a garden. Which feature confirms that it is an insect rather than an earthworm? (A) Bilateral symmetrical body (B) Cylindrical body (C) Body with little segmentation (D) Body with jointed legs Answer: Correct Option: (D) ► Why? D. Body with jointed legs is a feature that confirms an organism is an insect, distinguishing it from an earthworm which lacks jointed legs. [Ch 12: Patterns in Life: Diversity and Classification]	1
2	What kind of elements generally donate their valence electrons to achieve a stable electronic configuration? (A) Elements with exactly four valence electrons (B) Elements with less than four valence electrons in their valence shell (C) Elements with more than four valence electrons (D) Elements with eight valence electrons Answer: Correct Option: (B) ► Why? B. If the valence shell of an atom has less than four electrons, it would generally donate its valence electrons to achieve a stable electronic configuration. [Ch 9: Atomic Foundations of Matter]	1



Q.NO.	QUESTIONS	MARKS
3	A cart starts from rest with zero velocity and travels a distance s with acceleration a in time T. Which kinematic equation correctly relates these variables? (A) $s = aT$ (B) $s = 1/2aT^2$ (C) $s = 1/2aT$ (D) $s = aT^2$ Answer: Correct Option: (B) ► Why? B. For an object starting from rest, the kinematic equation relating distance, acceleration, and time is $s = 1/2aT^2$. [Ch 6: How Forces Affect Motion]	1
4	Flowers that bloom at night are generally white or light in colour compared to flowers that bloom during the day. What is the most likely reason for this? (A) Light colours help the flowers retain more water at night (B) Dark colours would cause the flowers to overheat during the night (C) Light colours make the flowers more visible to pollinators that are active at night (D) Light colours prevent the flowers from closing their petals at night Answer: Correct Option: (C) ► Why? C. Light colours make the flowers more visible to pollinators that are active at night. White or light-coloured flowers stand out in the dark, helping nocturnal pollinators locate them easily. [Ch 11: Reproduction: How Life Continues]	1
5	What is the process by which meristematic tissue becomes specialised to perform specific functions called? (A) Differentiation (B) Transpiration (C) Diffusion (D) Conduction Answer: Correct Option: (A) ► Why? A. Differentiation is the process by which meristematic tissue becomes specialised and loses the ability to divide, forming permanent tissues. [Ch 3: Tissues in Action]	1
6	Assertion (A): Sclerenchyma tissue provides hardness and strength to plants like the coconut husk and walnut shell. Reason (R): Sclerenchyma cells have thick walls due to the deposition of pectin. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.	1



Q.NO.	QUESTIONS	MARKS
	Answer: Correct Option: (C) ► Why? C. The assertion is true because sclerenchyma provides hardness and strength to structures like the coconut husk and walnut shell. The reason is false because sclerenchyma cells have thick walls due to the deposition of lignin, not pectin. [Ch 3: Tissues in Action]	
7	Assertion (A): Work is done by a force on an object when the force displaces the object in the direction of the force. Reason (R): An object having the capacity to do work is said to possess energy. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. Both A and R are true, but R is NOT the correct explanation of A. The assertion correctly defines the condition for work to be done, and the reason correctly defines energy, but the definition of energy does not explain why work requires displacement in the direction of the force. [Ch 7: Work, Energy, and Simple Machines]	1
8	Which pair of cell organelles contains DNA? (A) Mitochondria and Nucleus (B) Ribosomes and Lysosomes (C) Golgi bodies and Ribosomes (D) Nucleus and Lysosomes Answer: Correct Option: (A) ► Why? A. Mitochondria and the Nucleus are the pair of cell organelles that contain DNA. [Ch 2: Cell: The Building Block of Life]	1
Section B Attempt all 4 questions. Each carries 2 mark(s).		
9	How can one scientifically disprove the claim that food becomes harmful during an eclipse? Answer: An eclipse is just a play of shadows. Asking scientific questions reveals no physical, chemical, or biological mechanism supports the claim. Questions about measurable data—like whether temperature changes significantly during an eclipse, or whether food goes bad in a shadow—help disprove it. ► Why? This tests the application of scientific questioning to evaluate a viral claim, as illustrated in the source's eclipse example. [Ch 1: Exploration: Entering the World of Secondary Science]	2



Q.NO.	QUESTIONS	MARKS
10	If 25 cells are seen along the diameter of a visible field measuring 5000 μm, calculate the estimated size of one cell. Answer: Using the formula: Estimated size = Diameter of visible field / Number of cells along the diameter = $5000 \mu\text{m} / 25 = 200 \mu\text{m}$. The estimated size of one cell is $200 \mu\text{m}$. ► Why? This question tests the student's ability to apply the formula for estimating cell size using the diameter of the visible field and the number of cells along that diameter. [Ch 2: Cell: The Building Block of Life]	2
11	Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg^{2+}. However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride. Answer: Cl^-, one, two ► Why? According to the textbook context, a chlorine atom gains one electron to acquire a negative charge, becoming a chloride anion represented as Cl^-. Since a magnesium atom gives two electrons to become Mg^{2+}, and each chlorine atom can only accept one electron, it requires two chlorine atoms to accept the two electrons. Therefore, one ion of magnesium and two ions of chlorine combine to form magnesium chloride, ensuring that the positive and negative charges balance each other to create a neutral overall structure. ► Step-by-step: - Step 1: Identify the ion formed by chlorine. The context states chlorine gains one electron to become a chloride anion, Cl^-. - Step 2: Determine the ratio of ions. Magnesium forms a Mg^{2+} ion by giving two electrons. Since each Cl atom takes only one electron, two Cl^- ions are needed to accept the two electrons from one Mg^{2+} ion. - Step 3: Fill in the blanks with 'Cl^-', 'one', and 'two'. [Ch 9: Atomic Foundations of Matter]	2
12	Name the four components of xylem. Which one is the only living component? Answer: Xylem consists of four components: tracheids, vessels, xylem parenchyma, and xylem fibres. Xylem parenchyma is the only living component of xylem, while tracheids, vessels, and xylem fibres are primarily sclerenchymatous. ► Why? The source lists all four xylem components and explicitly identifies xylem parenchyma as the sole living element, with the other three being primarily sclerenchymatous (dead). [Ch 3: Tissues in Action]	2
Section C Attempt 3 out of 4.		
13	Describe the structure of the cell membrane as explained by the fluid-mosaic model.	3



Q.NO.	QUESTIONS	MARKS
	Answer: The cell membrane is extremely thin, about 7 to 10 nanometres thick, and is made up of lipids and proteins. According to the fluid-mosaic model, the membrane has a lipid bilayer with proteins embedded in it. The lipid bilayer consists of two layers of special fat molecules with water-attracting heads outwards and water-repelling tails inwards. The molecules can move sideways, flip, and rotate within the membrane, making it fluid. Since the molecules are arranged like tiles in a mosaic, it is called the 'mosaic' model. The embedded proteins act like gatekeepers, helping substances pass through. ► <i>Why?</i> The fluid-mosaic model describes the cell membrane as a flexible lipid bilayer with embedded proteins that can move, creating a mosaic pattern. ► Step-by-step: 1. Step 1: State the basic composition of the membrane (lipids and proteins) and its thickness. 2. Step 2: Describe the lipid bilayer structure and the fluid nature of the molecules. 3. Step 3: Explain the mosaic arrangement and the role of embedded proteins. [Ch 2: Cell: The Building Block of Life]	
	OR Many smartphones have an inbuilt accelerometer that can detect very small accelerations. Install an app, such as Phyphox (phyphox.org) and open 'Accelerometer' (without g). Note the readings when (i) the phone is on an outstretched palm, and (ii) the phone is kept on the floor. What differences do you observe? What does this tell you about motion and acceleration in real situations? (a) What differences do you observe between the readings when the phone is on an outstretched palm and when kept on the floor? Answer: (a) When the phone is on an outstretched palm, the accelerometer detects tiny involuntary movements (small fluctuations in readings). When the phone is kept on the floor, these fluctuations are significantly reduced or absent, showing more stable readings. (b) This tells us that in real situations, even when we think an object is stationary, there can be small accelerations due to involuntary movements. These tiny movements are significant enough to be detected by sensitive instruments and are even studied in medical research for conditions like movement disorders. It shows that perfect rest is difficult to achieve in practice. ► <i>Why?</i> The textbook context mentions that such tiny involuntary movements are studied in medical research for movement disorders. When the phone is on an outstretched palm, the human hand naturally has small tremors/involuntary movements that the sensitive accelerometer can detect. On the floor, these human-induced movements are absent. This demonstrates that in real situations, objects we consider 'at rest' may still experience small accelerations, and that acceleration isn't just about large, visible changes in motion but also includes these minute, involuntary movements. [Ch 4: Describing Motion Around Us]	



Q.NO.	QUESTIONS	MARKS
14	Explain the role of prediction in science. What do scientists do when their predictions do not match their observations? Answer: Prediction is a remarkable strength of science that allows us to anticipate what will happen under new or different conditions before or even without performing an experiment. These predictions are not guesses but reasoned expectations based on evidence and careful thinking. When predictions match observations, confidence in the underlying science grows. However, when they do not match, scientists do not ignore it. Instead, they re-examine their assumptions, models, or measurements. This process makes prediction a powerful tool that drives further exploration and a deeper understanding of the world, showing that science is always open to correction. ► Why? Predictions are reasoned expectations based on evidence; when they fail, scientists re-examine their assumptions and models rather than rejecting ideas based on opinion. ► Step-by-step: 1. Step 1: Define scientific prediction as reasoned expectations based on evidence, not guesses. 2. Step 2: State that matching predictions and observations increases confidence in the science. 3. Step 3: Explain that mismatched predictions lead scientists to re-examine their assumptions, models, or measurements. [Ch 1: Exploration: Entering the World of Secondary Science]	3
15	F. C. Steward grew phloem cells of a carrot in a nutrient medium with simple sugars and hormones, observing the formation of a complete plant. Explain the processes of dedifferentiation and redifferentiation that occurred, and state the term for this cellular ability. Answer: Dedifferentiation is the process where mature phloem cells regain the ability to divide, forming an undifferentiated mass of unspecialised cells. Redifferentiation occurs when this undifferentiated mass, under appropriate conditions with nutrients and growth chemicals, further divides and differentiates to form specialised structures like roots and shoots, eventually becoming a complete plant. This ability of mature plant cells to undifferentiate, divide, and redifferentiate into a new plant is known as totipotency. ► Why? The answer defines dedifferentiation and redifferentiation exactly as described in the Steward experiment, culminating in the definition of totipotency derived from the source text. ► Step-by-step: 1. Step 1: Identify dedifferentiation as the process where mature cells regain the ability to divide into an unspecialised mass. 2. Step 2: Identify redifferentiation as the process where the unspecialised mass forms specialised structures like roots and shoots. 3. Step 3: Conclude that the overall ability to undifferentiate, divide, and redifferentiate into a new plant is called totipotency.	3



Q.NO.	QUESTIONS	MARKS
	[Ch 3: Tissues in Action]	
Section D Attempt 2 out of 3.		
16	Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess. Answer: In science, laws, theories, and principles are terms with specific and distinct meanings that help organise our understanding of the world in a systematic way. A law usually describes a regular pattern observed in nature, often expressed using words or mathematical relationships. For example, Newton's laws of motion explain the jerk felt when a bus stops suddenly. Laws describe what happens based on repeated observations. A theory goes a step further than a law. It provides an explanation of why those patterns occur, usually based on evidence gathered over time and available at that time. For example, the atomic theory explains how molecules are formed. In science, a theory does not mean a guess or an untested idea; it is an explanation based on careful testing and critical examination. Principles are broad ideas that help us make sense of a given situation. For example, the principle of conservation of energy can be applied when climbing up the stairs. Principles serve as guiding ideas rather than descriptions or explanations. A key feature that makes science reliable is that these ideas are always open to improvement and often change as new evidence becomes available. No scientific theory is ever final and none is beyond question. This openness to being corrected by nature itself is what allows science to help us understand the world we live in. ► Why? Laws describe observed patterns, theories explain why patterns occur based on evidence, and principles are broad guiding ideas. A scientific theory is a well-tested explanation, not a guess. ► Step-by-step: 1. Step 1: Define a law and give the Newton's laws of motion example. 2. Step 2: Define a theory and give the atomic theory example; clarify it is not a guess. 3. Step 3: Define a principle and give the conservation of energy example. 4. Step 4: Explain how all these ideas remain open to improvement as new evidence emerges. 5. Step 5: Conclude by stating why this openness makes science reliable. [Ch 1: Exploration: Entering the World of Secondary Science]	5
	OR Explain how a microscope enables us to study cell structure. Describe the three main features that scientists have improved in microscopes over the years, and demonstrate how the estimated size of a cell and its total magnification are calculated.	



Q.NO.	QUESTIONS	MARKS
	Answer: A microscope allows us to see very small structures clearly and is an essential tool for studying cell structure. Over the years, scientists have turned the microscope into a powerful tool by improving its three main features. Resolution is a measure of clarity of the image, allowing us to distinguish closely spaced structures within a cell. Contrast refers to the difference in brightness between various parts of an object, helping to highlight different cellular components. Magnification is the ability to enlarge the image of a specimen, enabling us to observe details invisible to the naked eye. The estimated size of a cell can be determined using the formula: Estimated size of cell = Diameter of the visible field in micrometre / Number of cells along the diameter. For example, if 25 cells are seen along the diameter of a field of 5000 μm, the size of one cell = $5000 \mu\text{m} / 25 = 200 \mu\text{m}$. The total magnification of a microscope depends on the magnifying power of the eyepiece and the objective lens. If both the eyepiece and the objective lens have a magnifying power of 10X, then the total magnification = $10X \times 10X = 100X$. This means a cell with an estimated size of $200 \mu\text{m}$ will appear 100 times larger. These improvements in resolution, contrast, and magnification have collectively made the microscope a powerful and indispensable tool for studying cells. ► Why? <i>The microscope is essential for cell study because it enhances resolution, contrast, and magnification, and cell size can be quantified using the field diameter divided by the number of cells, while total magnification is the product of eyepiece and objective lens powers.</i> ► Step-by-step: 1. Step 1: State that a microscope is an essential tool for studying cell structure by making very small structures clearly visible. 2. Step 2: Describe the three improved features — resolution (clarity), contrast (brightness difference between parts), and magnification (enlargement of image). 3. Step 3: State the formula for estimated cell size and work through the example ($5000 \mu\text{m} / 25 \text{ cells} = 200 \mu\text{m}$). 4. Step 4: Explain total magnification as the product of eyepiece and objective lens magnifying powers ($10X \times 10X = 100X$). 5. Step 5: Conclude by summarising how improvements in all three features make microscopes powerful tools for cell study. <i>[Ch 2: Cell: The Building Block of Life]</i>	
17	Distinguish between voluntary and involuntary movements. Compare the structural features and functions of skeletal muscles, smooth muscles, and cardiac muscles.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Movements in the human body are classified as voluntary or involuntary based on conscious control. Voluntary movements, such as running, writing, or lifting objects, are under our conscious control and are carried out by skeletal muscles. Involuntary movements occur automatically without conscious control, for example the movement of food in the intestine and the beating of the heart. Skeletal muscles are attached to the skeleton and are made up of bundles of long, cylindrical cells called muscle fibres. These fibres are unbranched, multinucleate (having many nuclei), and striated (showing light and dark bands). They produce voluntary movements by pulling on bones. Smooth muscles are found in organs like the stomach and intestines. Their cells are spindle-shaped, have a single nucleus, and lack striations. They are responsible for slow, continuous involuntary movements such as digestion. Cardiac muscles are found only in the heart. Their fibres are cylindrical and branched with a single nucleus and have faint striations. They work tirelessly and rhythmically, enabling the heart to beat throughout life without fatigue. In summary, skeletal muscles are voluntary, striated, and multinucleate; smooth muscles are involuntary, non-striated, and uninucleate; and cardiac muscles are involuntary, faintly striated, branched, and uninucleate — each type uniquely adapted to its function. ► Why? <i>The three muscle types differ in their control mechanism (voluntary vs involuntary), cell shape, number of nuclei, presence of striations, and specific functions, reflecting their adaptation to different roles in the body.</i> ► Step-by-step: 1. Step 1: Define voluntary and involuntary movements with examples. 2. Step 2: Describe skeletal muscles — attached to skeleton, unbranched, multinucleate, striated, voluntary. 3. Step 3: Describe smooth muscles — in stomach/intestines, spindle-shaped, uninucleate, non-striated, involuntary, slow continuous movement. 4. Step 4: Describe cardiac muscles — only in heart, cylindrical and branched, uninucleate, faint striations, rhythmic and tireless. 5. Step 5: Summarise the key structural and functional differences among the three muscle types. <i>[Ch 3: Tissues in Action]</i>	
Section E Attempt 1 out of 1.		



Q.NO.	QUESTIONS	MARKS
18	Case Study: Building Scientific Models A student wants to model how long it takes for a ball thrown upward to return to the ground. She considers several factors: the mass of the ball, its initial speed and direction, air resistance, the colour of the ball, and the spin of the ball. She recalls that physicist Meghnad Saha simplified the study of stars by ignoring complex processes and focusing only on temperature, pressure, and how atoms formed ions, which helped explain the connection between star colour and temperature. She decides to apply a similar simplification approach. She also notes that science uses precise symbols like m for mass, v for velocity, and F for force, and that equations are compact statements about relationships between quantities. (i) What three factors did Meghnad Saha focus on to explain the colour of stars? (ii) Why should the student ignore the colour of the ball when modeling the time it takes to return to the ground? (iii) The student ignores air resistance and spin in her simple model. Later, she wants a more accurate prediction. What should she do, based on the approach described for building complex models? (iv) The student writes an equation relating distance, time, and velocity to describe the ball's motion. Explain why this equation is more than just a calculation tool. Answer: (i) Temperature, pressure, and how atoms formed ions. (ii) The colour of the ball does not affect the motion of the ball, just as the colour of a cricket ball makes no difference to whether it crosses the boundary. Only factors that affect the outcome should be included. (iii) She should add extra details such as air resistance and spin for greater accuracy. As we build more and more complex models, we add extra details beyond the simple model to improve the accuracy of our predictions. (iv) An equation is a compact statement about how certain things are related. Mathematics in science is a language that helps us think more clearly about the world, not just a tool for finding numerical answers. ► Why? i) Temperature, pressure, and how atoms formed ions. ii) The colour of the ball does not affect the motion of the ball, just as the colour of a cricket ball makes no difference to whether it crosses the boundary. Only factors that affect the outcome should be included. iii) She should add extra details such as air resistance and spin for greater accuracy. As we build more and more complex models, we add extra details beyond the simple model to improve the accuracy of our predictions. iv) An equation is a compact statement about how certain things are related. Mathematics in science is a language that helps us think more clearly about the world, not just a tool for finding numerical answers. [Ch 1: Exploration: Entering the World of Secondary Science]	5

